

Jiyu Hu

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Education

University of Illinois Urbana-Champaign <i>Ph.D. in Computer Science</i>	Aug. 2023 – Present Urbana, IL
Carnegie Mellon University – School of Computer Science <i>Master of Computational Data Science</i>	Aug. 2021 – May 2023 Pittsburgh, PA
University of Illinois Urbana-Champaign <i>Bachelor of Science in Computer Engineering</i>	Aug. 2017 – May 2021 Urbana, IL

Publications

* = Equal Contribution

SOSP '26	Disk-Based LSMs: An Unexpectedly Good Index for Partly Coherent CXL Memory Kiran Hombal, <u>Jiyu Hu</u> , Marcos K. Aguilera, Ramnatthan Alagappan, and Aishwarya Ganesan <i>ACM SIGOPS 32th Symposium on Operating Systems Principles, 2026</i>
OSDI '26	MEGALON: Efficient Data Sharing for Partly Coherent CXL Memory <u>Jiyu Hu</u> , Seokjoo Cho*, Landon Johnson*, Kiran Hombal, Shreesha G. Bhat, Marcos K. Aguilera, Ramnatthan Alagappan, and Aishwarya Ganesan <i>20th USENIX Symposium on Operating Systems Design and Implementation, 2026</i>
TOCS	LazyLog: A New Shared Log Abstraction and Design for Modern Low-Latency Applications Xuhao Luo, Shreesha G. Bhat*, <u>Jiyu Hu</u> *, Ramnatthan Alagappan, and Aishwarya Ganesan <i>ACM Transactions on Computer Systems, 2025</i>
OSDI '25	Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering Shreesha G. Bhat, Tony Hong, Xuhao Luo, <u>Jiyu Hu</u> , Aishwarya Ganesan, and Ramnatthan Alagappan <i>19th USENIX Symposium on Operating Systems Design and Implementation, 2025</i>
SOSP '24	LazyLog: A New Shared Log Abstraction for Low-Latency Applications Xuhao Luo, Shreesha G. Bhat*, <u>Jiyu Hu</u> *, Ramnatthan Alagappan, and Aishwarya Ganesan <i>ACM SIGOPS 30th Symposium on Operating Systems Principles, 2024</i>
HotStorage '24	Rethinking Erasure-Coding Libraries in the Age of Optimized Machine Learning Jiyu Hu, Jack Kosaian, and K. V. Rashmi <i>16th ACM Workshop on Hot Topics in Storage and File Systems, 2024</i>
TNSM	Medley: A Membership Service for IoT Networks Rui Yang, Jiangran Wang, Jiyu Hu, Shichu Zhu, Yifei Li, and Indranil Gupta <i>IEEE Transactions on Network and Service Management, 2022</i>

SEC **B2-Coupon: Efficient and Non-intrusive Mobile Coupon Distribution using Dual Bloom Filter**
Xueda Shen*, Jiyu Hu*, Yunqi Zhang, and Ian C. Quinn
IEEE/ACM Symposium on Edge Computing, 2020

Talks

Intro to CXL: Data Sharing on Partly Coherent CXL Memory <i>Bytedance</i>	Jul. 2026 <i>Bellevue, WA</i>
Data Sharing on Partly Coherent CXL Memory <i>Gray Systems Lab, Microsoft</i>	Jul. 2026 <i>Redmond, WA</i>
Rethinking Erasure Coding Libraries in the Age of Optimized Machine Learning <i>CMU PDL Retreat</i>	Nov. 2022 <i>Pittsburgh, PA</i>

Awards & Scholarships

Jay Lepreau Best Paper Award Nominee (MEGALON) <i>OSDI '26</i>	2026
Student Travel Grant <i>OSDI '25 & OSDI '26</i>	2025, 2026
Best Paper Award (LazyLog) <i>SOSP '24</i>	2024
Senior Design Instructor's Award (The Best Senior Design Award) <i>University of Illinois Urbana-Champaign</i>	2021
Bradley A. Simons Memorial Scholarship <i>University of Illinois Urbana-Champaign</i>	2019
Dean's List <i>University of Illinois Urbana-Champaign</i>	2017, 2018, 2019, 2020, 2021

Research Experience

Prism <i>DASSL, University of Illinois Urbana-Champaign</i>	Jun. 2025 – Jun. 2026 <i>Urbana, IL</i>
Investigate efficient index design on CXL memory under the assumption of partial hardware coherence.	
MEGALON <i>DASSL, University of Illinois Urbana-Champaign</i>	Apr. 2024 – Dec. 2025 <i>Urbana, IL</i>
Investigate the design of efficient sharing of CXL memory under the assumption of partial hardware coherence.	
SpecLog <i>DASSL, University of Illinois Urbana-Champaign</i>	Apr. 2024 – Dec. 2024 <i>Urbana, IL</i>
Design and implement a new shared log abstraction that significantly decreases e2e latency by speculation.	
LazyLog <i>DASSL, University of Illinois Urbana-Champaign</i>	Jan. 2024 – Jul. 2024 <i>Urbana, IL</i>
Design and implement a new shared log abstraction that significantly decreases append latency from state-of-the-art shared log implementations by delaying ordering the appended log entries.	
TVM-EC <i>TheSys Lab, Carnegie Mellon University</i>	Jan. 2022 – Jul. 2024 <i>Pittsburgh, PA</i>
Propose a new way of implementing high-performance erasure-coding libraries via machine learning libraries, reducing the effort to design and maintain erasure coding libraries.	

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| Medley
<i>Distributed Protocols Research Group, University of Illinois Urbana-Champaign</i> | Jan. 2020 – Jul. 2021
<i>Urbana, IL</i> |
| <ul style="list-style-type: none"> • Develop and evaluate a new IoT failure detection protocol that is aware of the spatial locality of physical nodes so as to decrease the overall communication overhead in an unstable network environment. | |
| B²-Coupon
<i>Prof. Dong Xuan's Research Group, The Ohio State University</i> | Nov. 2019 – Nov. 2020
<i>Columbus, OH</i> |
| <ul style="list-style-type: none"> • Design a better coupon distribution protocol for mobile devices. | |

Services

Artifact Evaluation Committee: CAIS '26, FAST '26, SOSP '25, OSDI '24, ATC '24

Shadow PC: EuroSys '26

Teaching

CS 498 Cloud Computing Applications <i>University of Illinois Urbana-Champaign</i>	TA 2026
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Industry Experience

Gray Systems Lab, Microsoft Corporation <i>Research Intern (Supervisor: Rathijit Sen)</i>	May 2026 – Aug. 2026
• Work on disaggregated memory for database systems.	
USM, Exadata, Oracle America, Inc. <i>Software Intern</i>	May 2022 – Aug. 2022
• Worked on the kernel log aggregation and analysis framework of Oracle distributed database.	
CUDA Math Library, NVIDIA Corporation <i>Software Intern</i>	June 2020 – Aug. 2020
• Analyzed the floating point error propagation in cuBLAS GEMM kernel.	